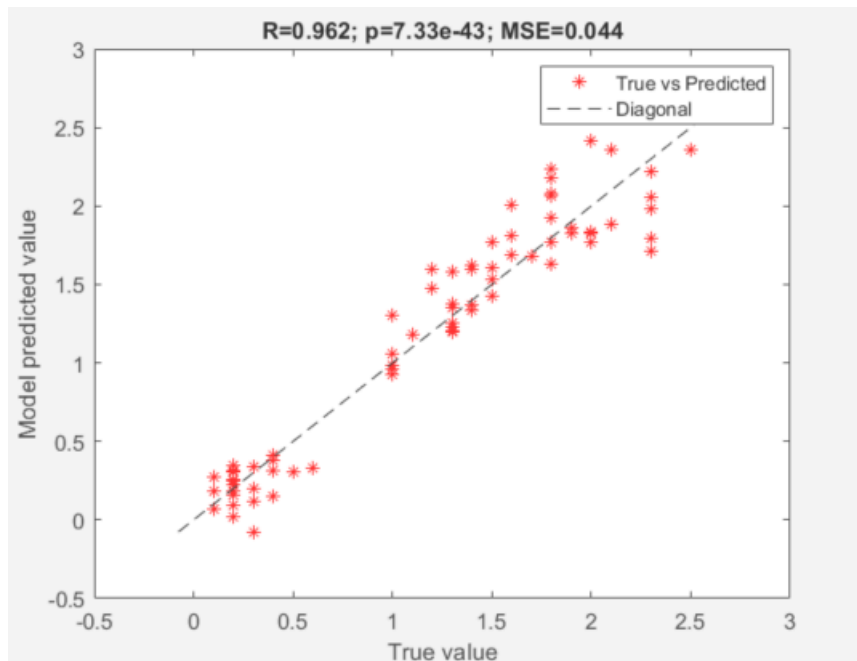


ErrorEstimationPlot

Create the **ErrorEstimationPlot** function that we created in the lecture. It takes as input an [Mx1] vector **prediction** and an [Mx1] vector **mse** between the **prediction** and the **target** vectors, the correlation **R** and statistical significance **p**-value from computing Pearson's correlation data **rg** used to draw the diagonal of the plot. It creates a plot in the current figure equivalent when running the example code.



Function ?

[Save](#)
[Reset](#)
[MATLAB Documentation \(https://www.mathworks.com/help/\)](https://www.mathworks.com/help/)

```

1 function [mse,R,p,rg] = EstimationErrorPlot(prediction,target)
2 mse = mean((prediction-target).^2);
3 plot(target, prediction, 'r*');
4 hold on;
5 xlabel('True Value');
6 ylabel('Model Predicted Value');
7 rg = [min([prediction; target]), max([prediction; target])];
8 plot(rg, rg, 'k--');
9 [R,p] = corr(target, prediction);
10 title(sprintf('R=%.3f; p=%.3g; MSE=%.3f', R, p, mse));
11 legend('True vs. Predicted', 'Diagonal');
12 end
13

```

Code to call your function ?

[Reset](#)

```

1 load fisheriris
2 D_train = meas(1:2:end,:);
3 D_test = meas(2:2:end,:);
4 mdl = fitlm(D_train(:,1:3),D_train(:,4));
5 prediction = predict(mdl,D_test(:,1:3));
6 [mse,R,p,rg] = EstimationErrorPlot(prediction,D_test(:,4));

```

[Run Function](#)


Assessment: All Tests Passed

Submit ?

✓ Is mse correct?

✓ Are R and p correct?

✓ Is rg correct?

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